

Harsh

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PROFESSIONAL SUMMARY

Diploma-level student in BS Data Science and Applications at the Indian Institute of Technology Madras, specializing in Fundamentals of Machine Learning, Generative AI, and scalable system architecture Proficient in building high-throughput data pipelines (1M+ records/day, 2+ PB scale), predictive models, ETL processes (Oracle SQL/PL-SQL), Tableau dashboards (which saved 10 hrs/week in manual reporting), and A/B testing (7.5% conversion uplift) Passionate about applying AI to railway infrastructure for predictive maintenance, smart track analytics, real-time digital reporting, and sustainable transformation aligned with Indian Railways, as well as innovations like Plasser Datamatic's contribution in delivering impactful, future-ready solutions.

EDUCATION

BN PUB SCH BADRI NARAIN MANDIR SECT 2 C FBD (40128)

Central Board of Secondary Education

2021 – 2023

Delhi, India

- 10th Grade: CGPA 8.1 / 10 | Top 1% who demonstrated exceptional performance in Literature achieving highest score among peers
- 12th Grade: CGPA 7.5 / 10 | Top 5% student to achieve highest marks in Informatics Practices and Data Structures and Algorithms

Indian Institute of Technology Madras

BS in Data Science and Applications: <https://ds.study.iitm.ac.in/student/23F3002742>

2023 – 2027 (expected)

Chennai, Tamil Nadu, India

- Learning: Machine Learning Reinforcement Learning Deep Learning Generative AI Big Data Analytics

SKILLS

Programming Languages: Python | C | CPP | CSharp | Rust | Java | JavaScript | R | Oracle | SQL | MySQL | PostgreSQL

Machine Learning: Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Scikit-learn

Deep Learning: Neural Networks, CNNs, RNNs, LSTMs, Transformers, TensorFlow, PyTorch

Reinforcement Learning: Markov Decision Processes (MDP), Q-Learning, Deep Q-Networks (DQN), Policy Gradients

Generative AI: GANs, Diffusion Models (conceptual), Large Language Models (LLMs), Prompt Engineering, Embeddings

Data & Big Data: Pandas, NumPy, Data Preprocessing, Spark, SQL, Analytics

MLOps & Deployment: Model Versioning, Experiment Tracking, Docker, Git, AWS (EC2, S3), REST APIs (basic)

Visualization & BI: Tableau, Power BI, Matplotlib, Seaborn, Polars, Grafana

CERTIFICATIONS

- HackerRank Skill Certificates: <https://github.com/nendroids/.certificates/tree/main/src/platforms/HackerRank>
- GeeksforGeeks Nation SkillUp Certificate: <https://github.com/nendroids/.certificates/tree/main/src/platforms/GeeksforGeeks>

RESEARCH

Probabilistic Modeling for Diabetes Diagnosis on Synthetic Clinical Data

Sep 2025 – Dec 2025

Kaggle: <https://www.kaggle.com/code/nendroids/pg-s05e12-diabetes-prediction-challenge>

Season 5 Episode 12

- Developed probabilistic machine learning models to predict diabetes diagnosis using synthetic tabular clinical data
- Performed data pre-processing feature engineering and model validation to improve predictive performance

RMSE-Optimized Predictive Modeling of Student Academic Performance

Jan 2026 – Feb 2026

Kaggle: <https://www.kaggle.com/code/nendroids/pg-s06e01-predicting-student-test-scores>

Season 6 Episode 01

- Developed supervised regression models to predict student exam scores using synthetically generated tabular educational data
- Optimized model performance using Root Mean Squared Error (RMSE) through feature engineering regularization and cross-validation

EXPERIENCE

Teaching Assistant (TA)

Sep 2025 – Dec 2025

Trainee

Chennai, Tamil Nadu, India

- Mentored 50+ students for programming in Python data structures and core algorithms with weekly hands-on coding sessions and one-on-one debugging support strengthening problem-solving logical reasoning and code optimization skills
- Designed and evaluated lab assignments and mini-projects focused on clean code algorithmic efficiency and real-world problem decomposition